

Data

Sample size and structure

Data files:
hw5_train_s9618.csv
hw5_test_s9618.csv

Training observations: 250

Test observations: 120

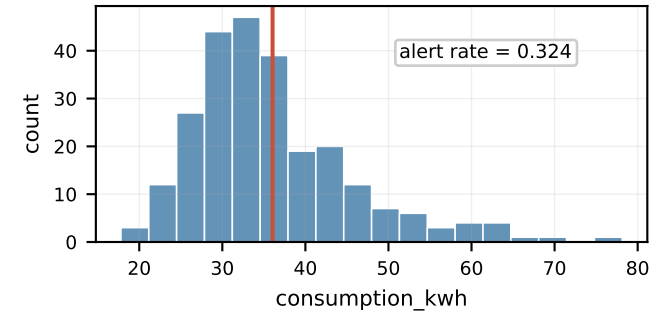
Covariates: temperature, humidity, weekend

Responses: consumption_kwh, high_demand_alert

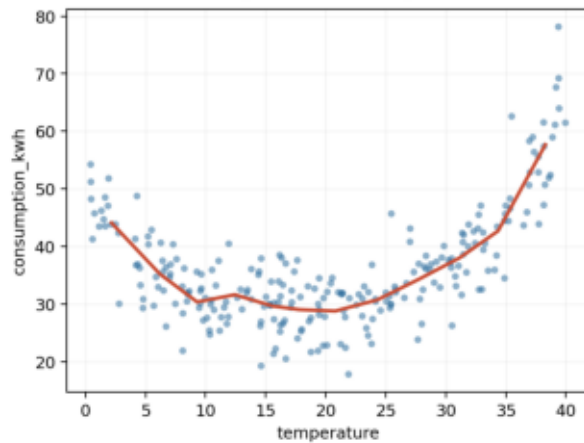
Y and covariate summary

variable	role	mean	sd	min	max
temp	covariate	20.00	11.08	0.44	39.96
humidity	covariate	54.03	20.56	20.47	89.97
weekend	covariate	0.34	0.48	0.00	1.00
consump	response	36.05	9.69	17.79	78.15
alert	response	0.32	0.47	0.00	1.00
test	hidden y	n=120		T 0.08	T 39.20

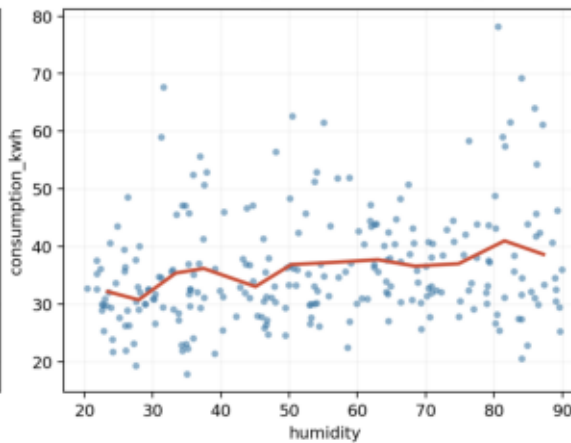
Response distribution



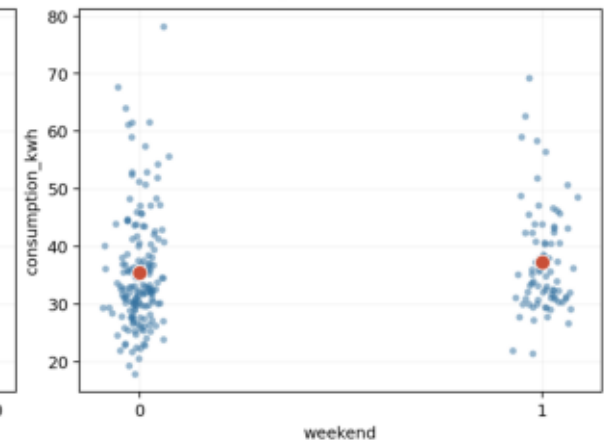
Consumption vs temperature



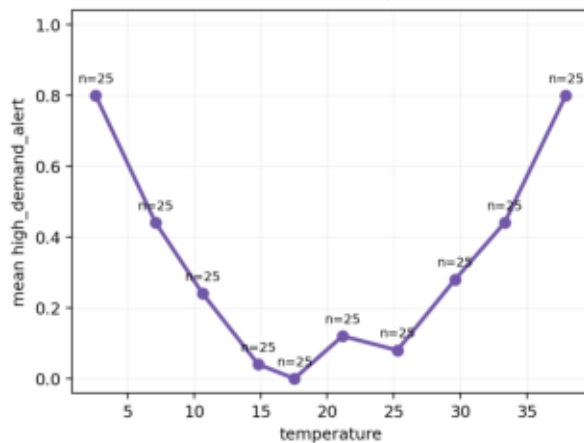
Consumption vs humidity



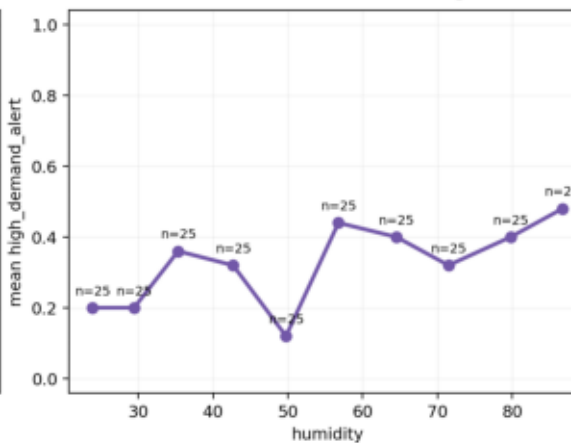
Consumption vs weekend



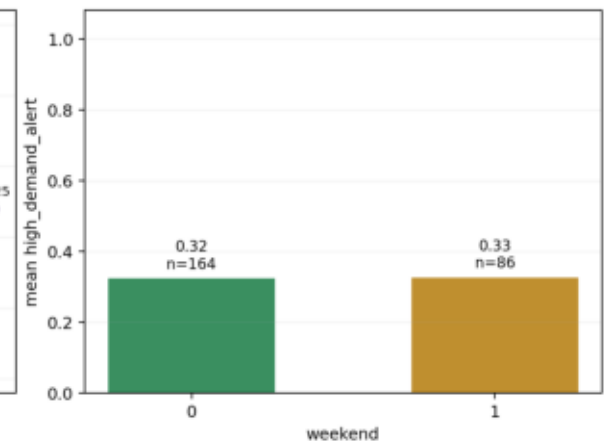
Alert rate vs temperature



Alert rate vs humidity



Alert rate vs weekend



Model, Method and Results

Model formula

Regression model:
 $y_i = \sum_j \beta_j B_j(T_i) + \beta_H H_i + \beta_W W_i + \epsilon_i$

Classification model:
 $\Pr(C_i=1) = \text{sigmoid}(\sum_j \beta_j B_j(T_i) + \beta_H H_i + \beta_W W_i)$

Estimated tuning:
regression internal knots: 2
classification internal knots: 0
boundary knots: 0 and 40 C
no intercept column

Method

Method:

1. Build clamped cubic knot vector
2. Construct B-spline basis manually by Cox-de Boor recursion
3. Fit OLS manually with `np.linalg.lstsq`
4. Fit no-intercept logistic regression on the hand-built design matrix
5. Choose knot count by manual 5-fold CV

Classification threshold: 0.5

Estimated regression coefficients

term	OLS beta
B1(T)	46.483
B2(T)	29.888
B3(T)	23.592
B4(T)	27.300
B5(T)	34.979
B6(T)	62.723
humidity	0.047
weekend	3.676

Estimated logistic coefficients

term	logit beta
B1(T)	2.405
B2(T)	-6.839
B3(T)	-4.865
B4(T)	1.204
humidity	0.019
weekend	0.473

Numerical results

quantity	value
regression CV MSE	20.4297
classification CV log loss	0.4608
classification CV Brier	0.1440
test consumption mean	37.146
test probability mean	0.403
predicted alert rate	0.425

Assumptions and libraries

Assumptions:

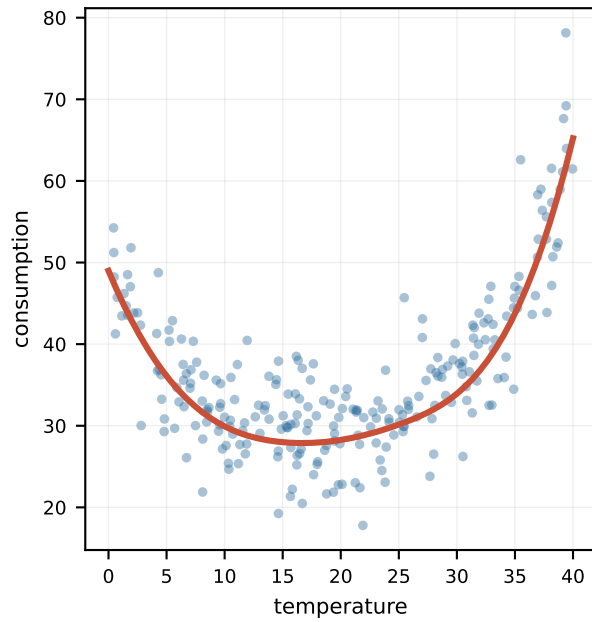
- independent rows, not a time series
- train and hidden test rows share the same process
- temperature effect is smooth but nonlinear
- humidity and weekend enter additively and linearly
- bootstrap resamples full rows

Libraries:

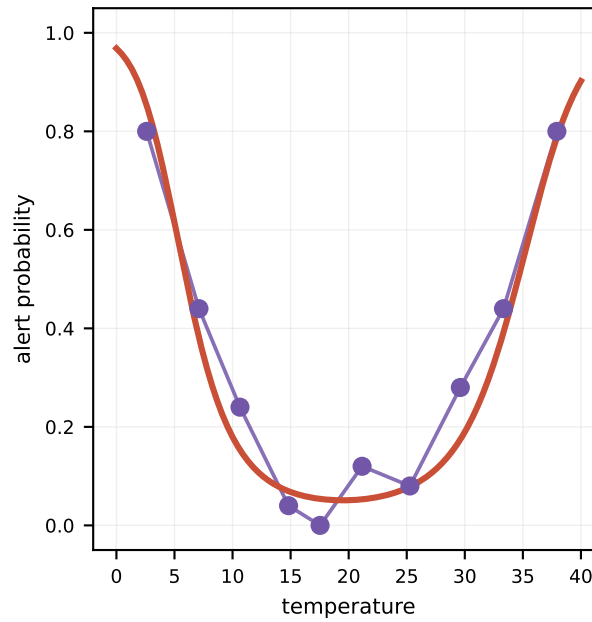
- `pandas`, `numpy`, `json`
- `sklearn.linear_model.LogisticRegression`
- `matplotlib` for slides/plots

Visualization

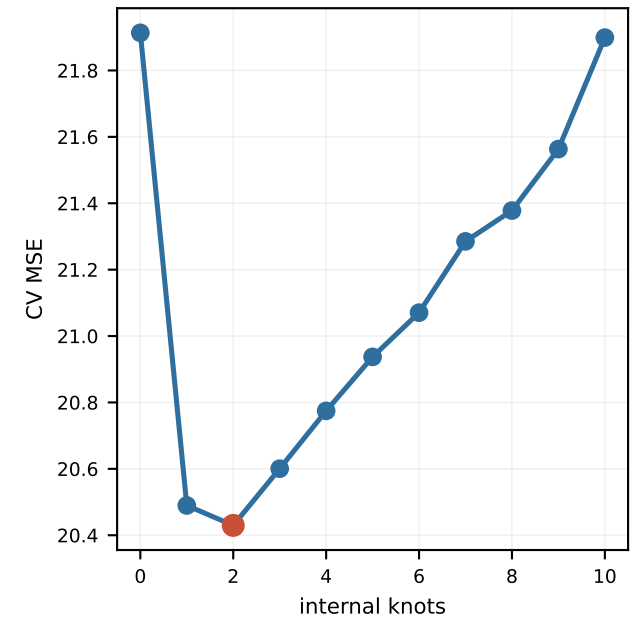
Consumption spline fit



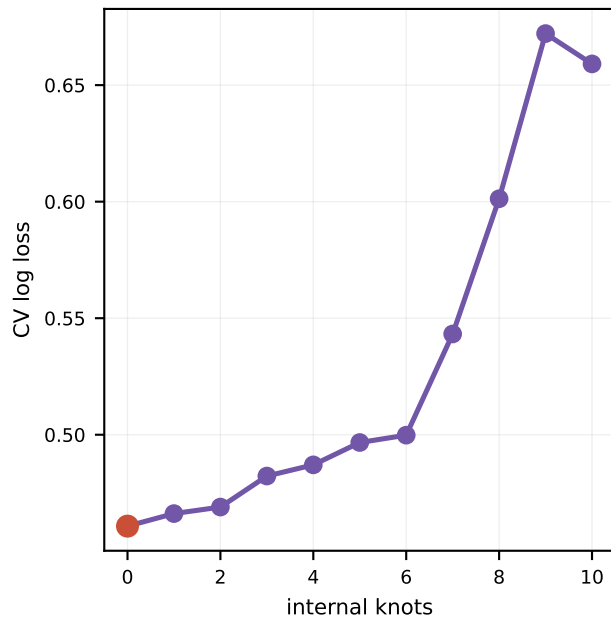
Alert probability spline fit



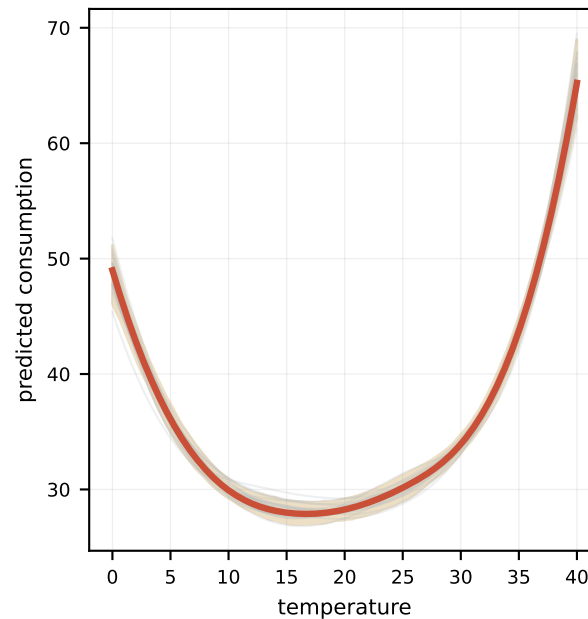
Regression CV error



Classifier CV log loss



Bootstrap stability: consumption



Bootstrap stability: alert prob

